RRB JE (CBT-2) — Electronics & Allied Engineering — Complete Weighted Book

Exam: RRB Junior Engineer, CEN No. 05/2025 (JE/DMS/CMA)

Exam Group: Electronics & Allied Engineering (Annexure-VII C)

Stage: CBT-2 (Second Stage) — 150 Questions / 150 Marks / 120 min / −⅓ negative marking

This book was built **weighted to the actual exam** — every chapter's depth is proportional to how heavily the topic is tested, derived from the official syllabus **and** a question-by-question analysis of the real 22-04-2025 CBT-2 Electronics paper. Every syllabus line item is covered; nothing is skipped.

How to use this book

- **Technical Abilities is 100 of 150 marks** — it is the whole ball game. Chapters 1–10 below cover it completely.
- The **three heaviest chapters** (Ch2 Devices & Circuits, Ch3 Digital Electronics, Ch7 Communication) carry ~16% each. Master these first.
- General Abilities (50 marks) is covered in the `general-abilities/` capsules, built on **NCERT** for Physics & Chemistry.
- For CBT-1 non-technical (Maths & Reasoning), the `pyq-research/` folder holds a curated PYQ bank instead of full theory, as requested.

Exam Pattern (CBT-2)

Section	Questions	Marks
General Awareness	15	15
Physics & Chemistry	15	15
Basics of Computer Applications	10	10
Basics of Environment & Pollution Control	10	10
Technical Abilities (Electronics)	**100**	**100**
Total	**150**	**150**

Negative marking: −1/3 mark per wrong answer.

Technical Abilities — Chapter Weightage

Ch	Chapter	Weight	Priority
2	Electronic Devices & Circuits	~16%	★★★
3	Digital Electronics	~16%	★★★
7	Communication Engineering	~16%	★★★
1	Electronic Components & Materials	~10%	★★
9	Computer Programming (C/C++, DS, DBMS)	~10%	★★
5	Microprocessor & Microcontroller (8085/8051)	~8%	★★
8	Data Communication & Networks	~8%	★★
10	Basic Electrical Engineering (+ Protection, Earthing)	~8%	★★
4	Linear Integrated Circuits (Op-amp, 555, PLL)	~4%	★
6	Electronic Measurements	~4%	★

Contents

Technical Abilities

1. Electronic Components & Materials
2. Electronic Devices & Circuits
3. Digital Electronics
4. Linear Integrated Circuits
5. Microprocessor & Microcontroller
6. Electronic Measurements
7. Communication Engineering
8. Data Communication & Networks
9. Computer Programming
10. Basic Electrical Engineering

General Abilities (NCERT-based)

- Physics (NCERT)
- Chemistry (NCERT)
- Basics of Computer Applications
- Environment & Pollution Control
- General Awareness Capsule

PYQ Research (CBT-1 non-technical)

- CBT-1 Mathematics & Reasoning PYQ Bank

Blueprint & methodology: see 00_Blueprint_and_Weightage.md.